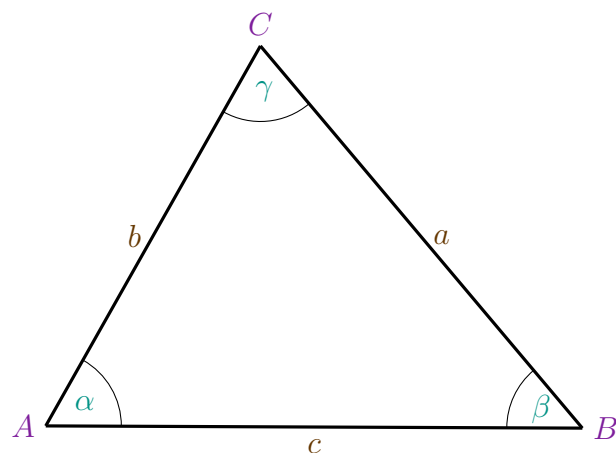
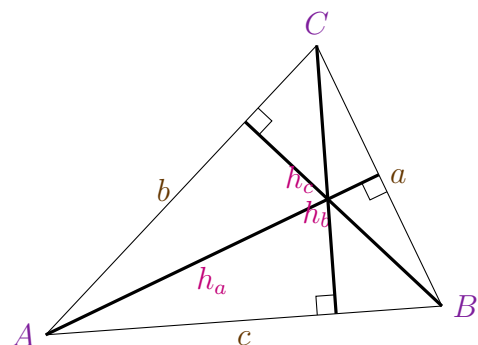


Reading suggestion: *How to Solve it* by George Pólya

Recall Question 6 from the session on Sunday (20 Sep 2026), where you constructed a triangle given three of its measurements. We will use the following notation in all the exercises that follow.



The vertices of the triangle are A , B , and C . The sides opposite the vertices A , B , and C are denoted by a , b , and c , respectively. The angles at A , B , and C are denoted by α , β , and γ , respectively.

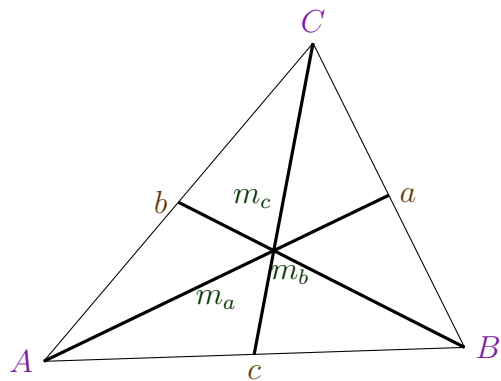


The altitude (or height measured from) from A to the side a (line segment BC) is denoted by h_a , and so on.

The median from A to the side a is m_a , and so on.

The angle bisector of α to the side a is d_α , and so on.

The radius of the circumscribed circle is R , and the radius of the inscribed circle is r .



In the following exercises, it may not always be possible to construct a triangle satisfying the given conditions. For example, try to construct a triangle with $a = 4$, $b = 5$, and $c = 10$. If you are unable to construct it, explain why the triangle is not possible.

Remember to break each problem down into smaller problems, as we did during the session. It may be helpful to write down your construction steps as you work.

Have fun constructing!

1. Construct a triangle given a , m_a , and b
2. Construct a triangle given a , m_a , and h_a
3. Construct a triangle given a , α , and h_a
4. Construct a triangle given a , α , and m_a
5. Construct a triangle given a , α , and r
6. Construct a triangle given a , c , and h_b
7. Construct a triangle given a , d_γ , and h_b
8. Construct a triangle given h_a , β , and h_b
9. Construct a triangle given α , β , and d_γ
10. Construct a triangle given h_a , h_b , and h_c
11. Construct a triangle given a , α , and R